

3DLOOK

FitXpress

Comprehensive Datasets
and Precision Metrics

May 2024



Data Foundations and Accuracy

Our AI has been trained on extensive data sets collected over eight years that deliver a high level of accuracy.

FitXpress is our solution that provides essential wellness related body composition data.

Data approach

3DLOOK's body-shape-centric data approach uses its patented AI powered mobile body scanning technology to create a 3D model and over 80 precise body measurements from just two smartphone photos.

Our solution, designed for scalability and flexibility, surpasses traditional hardware body scanners, delivering unparalleled reliability and adaptability.

Location:
US, Europe

Age Range:
16 to 78

Weight Range:
38 kg to 210 kg
(83.77 lbs) to (462.97 lbs)

Height Range:
150 cm to 205 cm
(5 ft 1 in) to (6 ft 9 in)

Study sample: **5000+**
participants

Additional Essential Data Collected:

- 150,000+ Photographs
- 30,000+ 3D Scans
- 430,000+ Individual Measurements

Gender Distribution:

48% male
52% female

Dataset Generation Protocol

By independently creating our datasets, we have achieved a highly detailed and accurate approach to generating final results, thereby providing extensive and precise body data.

Approach for data collection methodology

Measurement Process: Hardware Scanner

Parameters measured:
86 different parameters per person

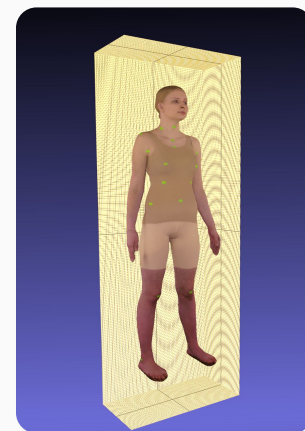
Includes measurements in sitting position and during breathing (inhale/exhale)

Allows study of the body in numerous configurations with maximum detail

Cameras used:
4 dynamic cameras

Additional Data Collected:

BMI, BMR, fat fold depth, body muscle, and body fat percentage using scales



Points Collected: over 5 million points for each 3D model

Ensures high accuracy

Preserves edges, shades and textures

Realistically transmits facial expressions, hair, and clothing

Photo Flow Simulation:

34 different photo configurations per user

Variations include distance, angle, slope, lighting

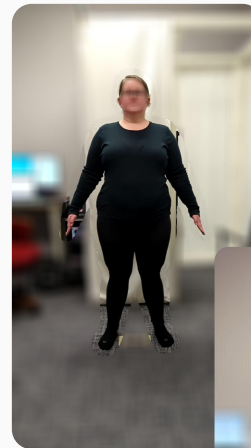


Collaborative Dataset Expansion

(North Carolina University)

This dataset creation was conducted in partnership with North Carolina State University/ Wilson College of Textiles to facilitate the enhancement of 3DLOOK's dataset and measurement accuracy.

Utilizing the University's Size Stream Body Scanner, comprehensive 3D body scans were obtained providing robust and precise datasets that contribute to the reliability and validity of our technology.



Assets used:

Size Stream Body Scanner

3D scans

Participant Photos

Front and Side

Casual/Sports clothes

Tight fitting athletic clothes

Metadata Collected

Height

Weight

Age

Gender

Ethnicity

Occupation

Body Fat Percentage Calculation

To accurately calculate body fat percentage, we utilize the U.S. Navy formula, a scientifically validated method based on precise anthropometric measurements. The robustness of these calculations is ensured by leveraging our complex algorithms and extensive datasets, providing a high degree of reliability and validity to the resulting body fat percentages.

The U.S. Navy formula for body percentage calculation:

For Men:

$$BF (\%) = 495 / (1.0324 - 0.19077 \times \log_{10} (AC-NC) + 0.15456 \times \log_{10} (H)) - 450$$
, where:

AC is Abdomen Circumference (cm)

NC is Neck Circumference (cm)

H is Height in cm

For Women:

$$BF (\%) = 495 / (1.29579 - 0.35004 \times \log_{10} (WC+HC-NC) + 0.22100 \times \log_{10} (H)) - 450$$
, where:

WC is Waist Circumference (cm)

HC is Hip Circumference (cm)

NC is Neck Circumference (cm)

H is Height in cm

Why We Chose the U.S. Navy formula for Body Fat Percentage Calculation

3DLOOK conducted a comprehensive evaluation of different body fat percentage calculation methods. Our research compared the Brozek and U.S. Navy formulas, given their widespread use and scientific validation.

The U.S. Navy formula was selected based on its superior accuracy, validated through rigorous testing on our extensive datasets.

Conclusion:

these comparative findings indicate that the U.S. Navy formula produces more precise estimations of body fat % within the parameters, thus establishing it as a more efficient approach in this particular context.

Comparison of Formulas:

Brozek formula:

Mean Absolute Error: 12.91%

Standard Deviation: 9.37%

vs

U.S. Navy Formula:

For Males:

Mean Absolute Error: 12.91%

Standard Deviation: 9.37%

For Females:

Mean Absolute Error: 12.91%

Standard Deviation: 9.37%

Fat Body Mass and Lean Body Mass Calculations

To accurately determine body composition, we use precise formulas to calculate fat body mass and lean body mass, providing valuable insights for personalized health and wellness strategies.

Fat Body Mass Calculations

User Defined Weight:

Weight (set by user) (kg or lbs) *
body fat % (calculated by
3DLOOK)"/100% = fat body mass (kg
or lbs)

Calculated Weight (Smart Scales):

Weight (smart scales) (kg or lbs) *
body fat % (calculated by
3DLOOK)"/100% = fat body mass (kg
or lbs)

Lean Body Mass Calculations

User Defined Weight:

Weight (set by user) kg or lbs - fat
body mass in kg or lbs = lean body
mass (kg or lbs)

Calculated Weight (Smart Scales):

Weight (smart scales (kg or lbs) - fat
body mass (kg or lbs) = lean body
mass in kg or lbs

BMI and BMR Calculations

Body Mass Index (BMI)

BMI= weight (kg) / height (m²)

Users input their height and weight, and we calculate their BMI using this formula.

If the weight is not provided, we use our weight prediction algorithm (smart scales beta) to estimate the weight for the calculation.

Adult BMI Categories:

< 16.0 - Severely Underweight
16.0 - 18.4 - Underweight
18.5 - 24.9 - Normal
25.0 - 29.9 - Overweight
30.0 - 34.9 - Moderately Obese
35.0 - 39.9 - Severely Obese
> 40.0 - Morbidly Obese

Basal Metabolic Rate (BMR)

BMR estimates the energy expended at rest in a neutrally temperate environment and in a post-absorptive state.

We calculate BMR using the Mifflin-St Jeor formula, requiring user weight, height, and age

BMR for Women=

$(10 * \text{weight [kg]}) + (6.25 * \text{height [cm]}) - (5 * \text{age [years]}) - 161$
BMR for Men=
 $(10 * \text{weight [kg]}) + (6.25 * \text{height [cm]}) - (5 * \text{age [years]}) + 5$

Smart Scales

We have developed a proprietary algorithm for determining body weight using two photos (front and side) and a person's height. This algorithm calculates the overall body volume as well as the volumes of individual body segments by summing the masses of these segments.

***Smart Scales are in beta as we continue to train and refine these algorithms

Algorithm Explanation:

The weight is estimated using the formula:

Weight = $\sum (V_i * \rho_i)$ for all body parts

Where:

V_i represents the volume of the i -th body part (derived from its mesh)

ρ_i is a predetermined coefficient representing "average density," calibrated from our ScanLab dataset

Error Analysis:

Based on our ScanLab test dataset, we conducted error analysis with the following results:

Mean Absolute Error: 2.1 kg

Standard Deviation of Absolute Error: 1.49 kg

Mean Relative Error: 3.11 %

Standard Deviation of Relative Error: 2.45 %



Thank You!